

CCNA 200-301 Notes

Chapter 23 – Extended Access Control List (Extended ACL)

Overview

Extended ACL filters traffic based on Source IP, Destination IP, Protocol, and Port Number. It is applied nearest to the source.

Standard vs Extended ACL

Standard ACL	Extended ACL
1-99	100-199
Source only	Source + Destination
No protocol/port	Protocol + Port
Near Destination	Near Source

Syntax

access-list NUMBER permit|deny protocol source wildcard destination wildcard [eq port]

Important Commands

- access-list 100 deny tcp host 30.0.0.2 host 20.0.0.100 eq 80
- access-list 100 permit ip any any
- access-list 102 deny tcp 200.200.200.0 0.0.0.255 host 192.168.1.50 eq 23
- access-list 105 deny tcp 200.200.200.0 0.0.0.255 host 192.168.1.50 eq 25
- access-list 110 deny udp 200.200.200.0 0.0.0.255 host 192.168.1.50 eq 53
- access-list 120 deny icmp 200.200.200.0 0.0.0.255 host 192.168.1.50 echo
- interface g0/0/1
- ip access-group 100 in
- ip access-group 100 out
- show access-lists
- show running-config
- show ip interface
- show ip interface brief
- show running-config | include access
- no access-list 100
- no ip access-group 100 in

Named ACL

```
ip access-list extended Sales
deny ip host 20.0.0.3 20.0.0.0 0.0.0.255
permit ip any any
interface f0/0
ip access-group Sales in
```

Common Ports

Port	Service
20	FTP
21	FTP Control
22	SSH
23	Telnet
25	SMTP
53	DNS
67/68	DHCP
69	TFTP
80	HTTP
110	POP3
161	SNMP
443	HTTPS

Lab Summary

Prevent host 30.0.0.2 from HTTP browsing to server 20.0.0.100 while allowing Ping and other traffic. Create the ACL, add 'permit ip any any', apply it nearest to the source, then verify using show commands.

Exam Notes

- Extended ACL range = 100-199
- Nearest to Source
- Supports TCP, UDP, ICMP, IP
- Uses eq for ports
- Implicit deny any
- Top-down processing

- First match wins
- Remember permit ip any any